The Canadian Urban Environmental Health Research Consortium

Canue Metadata - Weather Flood Susceptibility Index

2026-11-06

DATA SET INFORMATION

Dataset Code: WTFSI_AVA_YY

Description:

CANUE acquired the Flood Susceptibility Index (FSI) dataset from Natural Resources Canada. This national map of flood susceptibility or flood prone areas is based on patterns of historic flood events as predicted by an ensemble machine learning model. The native resolution of the FSI datasets is 30 m. CANUE obtained the FSI dataset for the HealthyPlan.City tool (https://healthyplan.city/en), which maps environmental and sociodemographic variables for 129 cities in Canada with a population of over 30,000 individuals. As such, this data is linked to 2021 postal codes only, as that is the target year of the HealthyPlan.City tool. Please note, however, that the FSI dataset here on canuedata.ca covers all of Canada (and not just the HealthyPlan.City footprint). Also note that the building-based exposure model developed by the University of Waterloo for HealthyPlan.City was not used for the data available here. For all 5 FSI variables, values close to 0 indicate low susceptibility to flooding and values close to 100 indicate areas that are the most susceptible to flooding.

Keywords: flood - natural hazard - hydrology - surface water - flood susceptibility

Place Keywords: Canada - National

GEOSPATIAL REFERENCE

Upper Left Corner: 65.14N , -141.02W

Lower Right Corner: 41.68N , -52.62W

Coordinate System: GCS_WGS84 - EPSG:4326

Geometry Type: POINT - Units: Decimal Degree

Geometry Data Source: DMTI Spatial Inc. (postal codes)

QUALITY ASSESSMENT

QA/QC Procedures:

CANUE did not assess the quality of the data. Users should review the documentation provided in the recommended citation and the supporting documentation listed.

Geographic Coordinate Positional Accuracy:

These metrics are linked to the corresponding annual postal codes files for mapping and analysis purposes. Refer to the postal code metadata file in Supporting Documentation for more information.

Vertical Positional Accuracy: N/A

Attribute Accuracy: N/A

Data Validity: NoData = -9999 (for numeric fields) - NoData=null (for category fields) - Data insufficient to calculate value = -1111

Associated Files: N/A

Data Comment:

DATA SOURCE

Data Source

Spatial Resolution:

Data Preparation Date: 2022-11-01

Beginning Date: 2021

End Date: 2021

Sampling Frequency of Data: Annual (varies)

Years Available:
2021

MAINTENANCE

Description: N/A

File Type: Comma separated values(.csv)

File Size: Individual year files range from 54 MB to 77 MB in size

Number of Data Files: 1

DATA USE CONDITIONS

The Data User is REQUIRED:
(i) to acknowledge data sources listed under Acknowledgement(s)
(ii) cite the publication(s) listed under Recommended Citation(s) as the providers and source of these data when using them in support of research, analysis, operations, policy decision or any other undertaking including publication(iii) complete and sign the CANUE Data Use and Sharing Agreement (available at http://canue.ca/data/), in which the name and signature of the researcher/analyst who takes responsibility for ensuring all conditions are met.

Data Sharing Restrictions:
These data files are provided solely for the purposes stated in the CANUE Data Sharing and Use Agreement and should not be re-distributed for any reason. These data also contain proprietary postal code data and may only be used for the project named in the CANUE Data Sharing and Use Agreement. Data can be shared only within a project team for the exclusive purposes of teaching, academic research and publishing, and/or planning of educational services in accordance to the DMTI End User Agreement associated with the Spatial Mapping Academic Research Tools (SMART) Program.

Include the following references in any publications resulting from the use of these data:
[1] McGrath, H. D. & Gohl, P. N. (2022). Accessing the impact of meteorological variables on machine learning flood susceptibility mapping. Remote Sensing, 14, 1-17. https://doi.org/10.3390/rs14071656

[2] McGrath, H.; Gohl, P.N. Prediction and Classification of Flood Susceptibility Based on Historic Record in a Large, Diverse, and Data Sparse Country. Environ. Sci. Proc. 2023, 25, 18. https://doi.org/10.3390/ECWS-7-14235

Include the following acknowledgements:
[1] Flood Susceptibility Index metrics, indexed to DMTI Spatial Inc. postal codes were provided by CANUE (Canadian Urban Environmental Health Research Consortium)

SUPPORT DOCUMENTATION

1 - Postal Code metadata (https://canuedata.ca/docs/CANUE-Browser-Metadata-PostalCodes.pdf)

2 - Link to FSI data on open.canada.ca (https://open.canada.ca/data/en/dataset/df106e11-4cee-425d-bd38-7e51ac674128)

VARIABLES

WTFSIYY_01 - Flood Susceptibility Index at Postal Code

Flood Susceptibility Index at Postal Code

WTFSIYY_02 - Flood Susceptibility Index within 100 m

Flood Susceptibility Index within 100 m

WTFSIYY_03 - Flood Susceptibility Index within 250 m

Flood Susceptibility Index within 250 m

WTFSIYY_04 - Flood Susceptibility Index within 500 m

Flood Susceptibility Index within 500 m

WTFSIYY_05 - Flood Susceptibility Index within 1000 m

Flood Susceptibility Index within 1000 m

SUPPORT CONTACT

Data Set Support Contact: info@canue.ca

Affiliated Organization:

CANUE (Canadian Urban Environmental Health Research Consortium) Dalla Lana School of Public Health, University of Toronto

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DATA SOURCE CONTACT

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National Resources Canada

- - Canada